Reflection Log

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Credit Name: CSE3910 - Project D
Assignment Name: Turn

How has your program changed from planning to coding to now?

The primary goal was to make the rover execute precise turns. While the initial plan was simple, several refinements were needed during testing to achieve accurate results.

1. **Imprecise Turn Angles**
   The rover's initial turn timing was too short, leading to incomplete rotations. I recalibrated the timing based on motor speed to achieve consistent angles.
   Code Example:

```
leftMotors.setTargetVelocity(-1);
rightMotors.setTargetVelocity(1);
Thread.sleep(750); // Adjusted timing for a precise 90-degree turn
```

2. **Stability During Pauses**
   After completing a turn, the motors sometimes kept moving slightly due to inertia. I added a brief pause to stabilize the rover.
   Code Example:

```
leftMotors.setTargetVelocity(0);
rightMotors.setTargetVelocity(0);
Thread.sleep(1000); // Pause to ensure stability
```

3. **Handling Uneven Surfaces**
   The rover occasionally veered off course on rough surfaces. To counter this, I reduced the motor speed during turns to improve control.
   Code Example:

```
leftMotors.setTargetVelocity(-0.5); // Reduced speed for more control
rightMotors.setTargetVelocity(0.5);
```

4. **Simplified Direction Changes**
   Initially, I implemented complex logic for alternating turns, but I simplified it by reusing consistent motor commands.
   Code Example:

```
leftMotors.setTargetVelocity(1);
rightMotors.setTargetVelocity(-1);
Thread.sleep(750); // Reusable logic for alternating turns
```

Final Thoughts: These refinements made the program more reliable and adaptable, ensuring that the rover could execute turns precisely and handle uneven conditions effectively.